10

SSH & Remote Profiles

A 4-hour session — lecture, live demonstrations, and hands-on exercises

Contents

Background

Introduction

This lesson covers secure remote connectivity in IBM Sterling B2B Integrator end to end: why SSH replaces plain FTP for trading-partner exchange, the four SSH keys the product uses, how a Perimeter Server bridges the DMZ, and how a Remote Profile packages a partner connection into a single, reusable definition. You will configure a profile with both password and key-based authentication, build and run a business process that moves a file over it, and practice diagnosing a connection failure from the logs.

Session format

This is a 4-hour session combining lecture, three live instructor demonstrations, and four hands-on exercises. A detailed minute-by-minute agenda is on slide 2 of the accompanying slide deck; use it to pace the room, but treat it as a guide rather than a strict script — let questions and lab pace adjust the timing live.

Lesson objectives

This lesson is designed to help you to:

- Explain why SSH/SFTP is preferred over plain FTP for B2B file transfer, and where it sits among the other protocols Sterling B2B Integrator supports
- Describe the SSH handshake, and the difference between symmetric and asymmetric encryption
- Identify all four SSH keys used in Sterling B2B Integrator, what each identifies, and where each is configured
- Explain the role of a Perimeter Server in a DMZ deployment
- Configure a Remote Profile, including authentication method and key reference, using the real product field set
- Build and execute a business process that uses the SFTP Client services to move a file through a Remote Profile
- Diagnose a failed secure connection using the correct log files, and apply baseline security best practices

Instructor Note

Open with the quick recap check on Exception Handling / Service Management before starting new content — see the accompanying slide deck, slide 3, for the two check questions. If the class needs a refresher, budget 10 extra minutes; otherwise proceed directly. A content-sourcing note: sections and exercises drawing directly on the IBM course guides or IBM product documentation are marked SOURCED with a citation; sections built from established public-domain SSH/SFTP protocol knowledge are marked GENERAL KNOWLEDGE. Both are legitimate course content — the tagging is there so you know which claims to defend from an official document versus general security-industry practice if a participant asks.

Why SSH? Securing Trading-Partner File Transfer

The problem

Plain FTP sends both credentials and file contents in clear text. On a shared or untrusted network, that traffic can be captured — a serious concern for EDI documents that routinely carry order, invoice, or shipment data, and often personally identifiable information. Plain FTP also gives the client no way to verify it is really talking to the intended server.

The SSH solution

SSH (Secure Shell) establishes an encrypted channel before any authentication or data transfer happens. SFTP (SSH File Transfer Protocol) runs file-transfer commands over that encrypted channel — it is a different protocol from FTPS, which is plain FTP wrapped in TLS/SSL rather than SSH. In Sterling B2B Integrator, SSH-based transfer is the default recommendation for any partner connection carrying business documents.

Plain FTP	SSH / SFTP
Credentials and data sent in clear text	Encrypted channel for both authentication and data
Password-only authentication	Supports key-based authentication
No server identity verification	Host-key trust prevents impersonation

The Secure File-Transfer Protocol Landscape

Why this matters SSH/SFTP is one of several protocols Sterling B2B Integrator and Sterling File Gateway support for partner connectivity. Knowing where SFTP sits relative to the alternatives helps you answer a very common real-world question: "why are we using SFTP for this partner and Connect:Direct for that one?"

Protocol	Transport security	Typical use
FTP	None	Legacy / internal only — avoid for partner-facing traffic
FTPS	TLS/SSL over FTP	FTP with certificate-based encryption
SFTP	SSH (encrypted channel + key or password auth)	Most common secure choice for B2B partner exchange
SCP	SSH	Simple secure copy; fewer directory operations than SFTP
Sterling Connect:Direct	Proprietary, point-to-point	High-volume, guaranteed-delivery enterprise transport
HTTP / HTTPS	TLS/SSL (HTTPS)	Web-services style integration and AS2-based exchange

Source: Managing Sterling File Gateway V2.6.6.1 (6F86G) — supported partner communication protocols (FTP, FTPS, SFTP, SCP, Sterling Connect:Direct, HTTP, HTTPS).

How SSH Works: The Handshake

General knowledge

This section applies to SSH in general — the same mechanics underlie any SSH/SFTP implementation, Sterling B2B Integrator included.

The six-step handshake

1. TCP connect — the client opens a TCP connection to the server, typically on port 22.
2. Version & algorithm negotiation — both sides agree on the SSH protocol version and which encryption, MAC, and key-exchange algorithms to use.
3. Key exchange (Diffie-Hellman) — client and server derive a shared symmetric session key without ever transmitting it. This key encrypts everything from this point on.
4. Server authentication (host key) — the server proves its identity by signing the exchange with its private host key; the client checks that signature against a trusted known-hosts entry.
5. Client authentication — the client authenticates by password, or by signing a challenge with its private key (public-key authentication).
6. Secure channel established — all subsequent traffic (directory listings, file contents, commands) flows through the encrypted channel; this is what SFTP rides on.

Crypto primer

Symmetric encryption uses one shared secret key to both encrypt and decrypt — fast, and used for the bulk data once a session is established, but it requires the two parties to agree on the secret without ever sending it in the clear. Asymmetric (public-key) encryption uses a mathematically-linked key pair: what one key encrypts, only the other can decrypt, and the private key never leaves its owner. SSH uses asymmetric cryptography for identity — proving "I hold the private key" by signing a challenge, without ever revealing the private key itself — and symmetric cryptography (via the Diffie-Hellman-derived session key) for the actual bulk data transfer, because it is far faster than asymmetric encryption at scale.

The Four SSH Keys in Sterling B2B Integrator

Overview Sterling B2B Integrator organizes SSH keys into two categories: two keys identify your trading partner, and two identify your own organization. Understanding which is which is the single most useful mental model for this entire lesson.

Known Host Key The public portion of your partner's remote SFTP Server adapter. You either receive it from the partner directly, or retrieve it automatically using Sterling B2B Integrator's SSH key-grabber utility. Configured in the Known Host Key section of the console. Identifies: your partner's server. Answers the question: "who am I connecting to?"

Authorized User Key Optional. The public key that identifies your partner when they connect into your Sterling B2B Integrator SFTP Server. Configured by checking in the key file the partner provides, then linking it to their user account under Accounts > [user] > Select Keys. Identifies: your partner's user account. Answers the question: "who is allowed to log in to me?"

User Identity Key Created within Sterling B2B Integrator, or generated externally (e.g., with OpenSSH) and imported. The public portion is checked out and shared with your trading partner. Referenced in an SSH Remote Profile, or as the UserIdentityKeyId BPML parameter, whenever your organization connects out to a partner. Identifies: your organization, outbound. Answers the question: "how do I prove it's me, connecting out?"

SSH Host Identity Key Goes on your SFTP Server adapter — used whenever your organization hosts an SFTP server for partners to connect into. Create it in Sterling B2B Integrator under Deployment > SSH Host Identity Key, check out its public portion for partners, and associate it with the SFTP Server adapter under Deployment > Services > Configuration. Identifies: your organization's server. Answers the question: "how does a partner know it's really my server hosting this?"

Key	Identifies	Configured where
Known Host Key	Partner's SFTP server	Known Host Key — manual check-in or automatic retrieval
Authorized User Key	Partner's user account (optional)	Check in key, link via Accounts > [user] > Select Keys
User Identity Key	Your organization, outbound	Create/import; referenced by SSH Remote Profile
SSH Host Identity Key	Your organization's server	Deployment > SSH Host Identity Key

Source: IBM Support — "The difference between the 4 SFTP (SSH) keys in IBM Sterling B2B Integrator"; menu path for SSH Host Identity Key confirmed in Sterling B2B Integrator Fundamentals (6F870), Deployment menu description.

Perimeter Servers

What it is

A perimeter server is an optional software tool for communications management, installed in a Demilitarized Zone (DMZ) — a computer host or small network inserted as a neutral zone between a company's private network and their public network. It is communication management software that manages communication flows between the perimeter network and Sterling B2B Integrator's TCP-based transport adapters.

Why use one

- Minimizes DMZ issues by keeping the application server out of the public-facing network path
- Enhances scalability and improves handling of large files
- Improves overall performance
- The perimeter server client inside Sterling B2B Integrator initiates contact and transmits files to/from the remote perimeter server over the internal LAN — the application server never needs a direct inbound port opened from the public network

Architecture

In a typical single-node deployment: trading partners communicate with the Remote Perimeter Server in the DMZ, usually over a WAN link. The perimeter server client in the Sterling B2B Integrator Server initiates contact and transmits files to/from the Remote PS over a LAN connection. The database server sits on a high-speed LAN link to the Sterling B2B Integrator Server, which also communicates with internal applications over LAN or WAN.

Component	Interaction
Perimeter Server (PS)	Receives a file from the trading partner. DMZ communications-management software that reduces network congestion, adds scalability, and enhances security.
Adapter	Receives the file (e.g., FTP, SFTP) and places the data into an initial workflow context, then starts the Workflow Engine.
Workflow Engine (WFE)	Loads the business process definition, determines the first service, and places the workflow context and step into one of the system queues.
Queue / Listener	One of the queues receives the data and business process to run; the listener attached to the queue runs the Activity Engine.
Activity Engine (AE)	Calls the service, receives results, requests a database connection from the pool, persists changes, and starts the next step.
Database	The repository for defined assets (business processes, maps) and most data; large documents can optionally be stored on the file system instead.

Source: IBM Sterling B2B Integrator Administration and Tuning (6F89G), Unit 1 — "High-Level Sterling B2B Integrator Architecture" and "Components of Sterling B2B Integrator" (component workflow scenario).

Remote Profiles: Overview and Configuration

Introduction	A Remote Profile is a named, reusable definition of a connection to a remote server. Instead of hard-coding a host name, port, and credentials into every business process, you configure them once as a Remote Profile, then reference that profile by name from any adapter that needs it.
Where it fits	A Business Process step (typically an SFTP Client Adapter) references a Remote Profile, which in turn defines how to reach the Partner Server. Changing the partner's connection details later means updating one profile, not every business process that uses it.
Menu path	**Trading Partners > SSH > Remote Profiles**

Field	Purpose
Remote Host	Partner's SFTP server address (accepts alphanumeric and dash characters only)
Remote User	Login user ID on the remote server (accepts alphanumeric, dash, underscore, and "$")
Port Number	Server's listening port — typically 22
Authentication Type	Password, or Public Key
SSH Password	Used when Authentication Type = Password (maximum length 55 characters)
Known Host Key	Selected from checked-in keys — with V5.2.5 and higher, multiple Known Host Keys can be checked in
User Identity Key	Selected when Authentication Type = Public Key
Connection Retry Count / Retry Delay	Number of retry attempts, and the delay in seconds between retries
Response Timeout	How long to wait for a server response, in seconds
Local Port Range / Compression	Local port allocation, and protocol compression setting

Source: IBM Sterling B2B Integrator product documentation — SSH Remote Profile field reference (menu path Trading Partners > SSH > Remote Profiles).

Getting the partner's Known Host Key

Automatic retrieval: Sterling B2B Integrator can connect to the remote host and collect the key for you, given the remote host address and (temporary) credentials — fastest in a lab/non-production setting.

Manual import: check in a key file the partner sends you directly, out of band. Supports SECSH and OpenSSH key file formats — the safer choice for production, since you are not trusting the network to hand you the key on first contact.

Source: IBM Sterling B2B Integrator product documentation — Known Host Key check-in methods.

Exercise 10-1: Create an SSH Remote Profile *(≈30 min, password authentication)*

Instructions

1. Navigate to Trading Partners > SSH > Remote Profiles.
2. Click Add, and provide a Remote Host name — use your assigned partner name as the profile identifier (e.g., ACMEFREIGHT_SFTP).
3. Enter Remote User, and the Port Number for the target SFTP server (default 22).
4. Set Authentication Type to Password for this first pass, and supply the test SSH Password provided in your lab environment.
5. Check in the partner's Known Host Key manually, or use automatic retrieval if your lab environment supports it.
6. Accept the default Connection Retry Count, Retry Delay, and Response Timeout values for now.
7. Save the profile, then use its Test / Validate action (or a simple Put) to confirm connectivity.

Verify: the profile shows a successful test connection before continuing to Exercise 10-2. Capture a screenshot of the successful result for your lab notes.

Key-Based Authentication

Why switch

Password authentication still transmits a secret on every connection and depends on that secret staying confidential. Key-based authentication removes the password entirely: the server verifies a cryptographic signature instead, using the four-key model covered earlier in this lesson.

Server-side setup

1. Create the Host Identity Key: Deployment > SSH Host Identity Key — generate the key your SFTP server presents to every connecting partner.
2. Configure the SFTP Server Adapter: Deployment > Services > Configuration — set Authentication to Password or Public Key, note the listening port, and enable the service.
3. Export the server's public key: check out the Host Identity Key's public portion to share with the client side.

Client-side setup

1. Create a Known Host Key on the client side by checking in the exported server public key.
2. Generate a User Identity Key on the client side (create or import a key pair), then check out its public portion.
3. Register the client's public key on the server: check in the User Identity Key's public portion as an Authorized User Key, and link it to the connecting user account under Accounts.

Configuring the profile

Edit the Remote Profile from Exercise 10-1: change Authentication Type to Public Key, and select the User Identity Key you just created. Confirm the Known Host Key entry is already in place from Exercise 10-1.

Source: IBM Sterling B2B Integrator product documentation, and IBM community reference material on SSH/SFTP key-based authentication setup in Sterling B2B Integrator.

Exercise 10-2: Switch to Key-Based Authentication *(≈30 min)*

Instructions

1. Generate a new User Identity Key pair (or import the key pair supplied for this lab).
2. Check out the public key value; this represents what you would send to a real trading partner out of band.
3. Check in the public key as an Authorized User Key and link it to the lab user account under Accounts > [user] > Select Keys.
4. Reopen the Remote Profile from Exercise 10-1 and change Authentication Type to Public Key, selecting the new User Identity Key.
5. Confirm the Known Host Key entry is still valid, or refresh it if the target server changed.
6. Re-test the connection and confirm it succeeds without a password prompt.

Verify: the same test connection you ran in Exercise 10-1 now succeeds using only the key pair — no SSH Password value is used.

SFTP Client Services and Building a Transfer

Introduction The FTP/SFTP Client Adapter uses a set of predefined client services to communicate with a remote server outside Sterling B2B Integrator. Each service performs one specific function and is invoked as a step inside a business process.

Service	What it does
Begin Session	Starts a session with the external trading partner to exchange documents
CD	Changes directory on the remote server
GET	Retrieves one or more documents from the remote directory into Sterling B2B Integrator
PUT	Places a document (or documents) into a directory on the remote server
LIST	Retrieves a list of documents in a specified remote directory
MOVE	Renames or moves a document from one remote directory to another
DELETE	Removes a document from the remote system — typically used after a successful GET, to prevent the file being pulled multiple times
PWD	Prints the current working directory on the remote server
End Session	Ends the session — used as the last step, and only after a Begin Session step has run

Source: Advanced Business Process Modeling (6F100G) — FTP/SFTP Client Services.

The business process for this lesson	Begin Session → PUT (sample outbound file) → GET (partner's acknowledgment) → End Session. Each step references the Remote Profile built in Exercises 10-1/10-2 by name — no host, port, or credentials are hard-coded into the business process itself.
Talking point	*The output file name you build in this exercise — for example ACME_SHIP_20260829.txt — is assembled from process data: partner code + document type + date. That is the simplest possible example of a field being mapped into an output. When EDI mapping is covered later in the course, the same field-to-field logic applies, just against much richer source and target structures.*

Exercise 10-3: Build and Run a Secure Transfer *(≈35 min)*

Instructions

1. Open the Graphical Process Modeler and create a new business process.
2. Add an SFTP Client Begin Session step configured to use your Remote Profile from Exercise 10-2 (key-based authentication).
3. Add a PUT step to send a sample outbound file to the partner directory.
4. Add a GET step to retrieve an acknowledgment file back from the partner.
5. Add an End Session step to close the connection.
6. Save, validate, and check in the business process, then execute it with the sample file provided.
7. Open Business Process > Monitor > Current Processes and confirm the instance completed successfully.

Verify: the instance shows a Complete state, and the acknowledgment file is visible in your local working directory.

Troubleshooting: Logs and Common Errors

Where to look Operations > System > Logs. Every adapter and service that participates in a secure transfer writes to its own log file — knowing which log matches which symptom is the fastest way to a fix.

Log	File name	What it tells you
SFTP Client Adapter and Services	sftpclient.log	Outbound SFTP connection and transfer activity from your business processes
SFTP Server Adapter	sftpserver.log	Inbound connections to your Sterling B2B Integrator SFTP server
SFTP Common Log	common3splogger.log	SFTP security errors — often the first place to check for authentication failures
FTP Client Adapter and Services	ftpclient.log	Outbound plain-FTP activity
FTP Server	ftp.log	Inbound plain-FTP activity
Perimeter Services	Perimeter.log	Perimeter server communication activity, when one is in the path
Security	security.log	Startup and component-licensing problems, including some authentication issues

Source: IBM Sterling B2B Integrator Administration and Tuning (6F89G) — Troubleshoot System Logs / Log File Types.

Common errors

"Host key not accepted" — the server's host key doesn't match what's in your Known Host Key list. Either the server was rebuilt/re-keyed, or this is a genuine impersonation attempt; verify out of band with the partner before re-trusting. Check sftpclient.log / common3splogger.log.

Authentication failure with keys configured — the public key isn't linked to the correct user account as an Authorized User Key, or the wrong User Identity Key is selected in the profile. Check sftpserver.log / security.log.

Connection times out — the firewall/DMZ path is blocked, or no Perimeter Server is configured where one is required. Check sftpclient.log / Perimeter.log.

Works from the command line, fails from Sterling B2B Integrator — compare the Remote Profile's Port / Host / Authentication Type against the working manual connection; a mismatched field is the usual culprit. Check sftpclient.log.

Changing a logging level

1. From the Administration Menu, select Operations > System > Logs.
2. Locate the relevant log (e.g., FTP Server Log, SFTP Client log) and click the pencil icon to its left.
3. Select a Logging Level — All, or Communication Trace — from the drop-down list, and click Save.
4. Reproduce the failure, then open the newest log file and review the additional detail.

Source: IBM Sterling B2B Integrator Administration and Tuning (6F89G), Exercise 4.2.1 / 4.2.2 — Reading the FTP Server Adapter / Accessing FTP Server Through Secured Layer (adapted here from FTP Server Adapter logging steps, which apply the same way to the SFTP equivalents).

Exercise 10-4: Server-Side Security and Diagnostics *(≈25 min)*

Instructions

1. From the Administration Menu, select Deployment > Services > Configuration; search by Service Type for the FTP Server Adapter (or SFTP Server Adapter, if available in your lab).
2. Edit the adapter and step through its configuration pages to the security page. Set SSL to Must, Cipher Strength to STRONG, Key Certificate Passphrase and Key Certificate (System Store) per your lab handout, and Clear Command Channel to Disallow.
3. Save and enable the service.
4. From a client that is NOT configured for the required security (a plain command-line client without SSL/matching keys), attempt a connection to the adapter's port and observe the failure.
5. Raise the relevant log's Logging Level to All (or Communication Trace) and repeat the failed attempt.
6. Open the newest log file and identify the specific error that explains the failure.
7. Correct the client-side configuration so that it matches the server's security requirements, and confirm the connection now succeeds.

Verify: you can point to the exact log line that explains the original failure, and the corrected client connects successfully.

Source: IBM Sterling B2B Integrator Administration and Tuning (6F89G), Exercise 4.2.1 / 4.2.2 — FTP Server Adapter SSL configuration fields and log-review procedure, adapted to a secure-connectivity diagnostic exercise.

Resource Portability and Security Best Practices

Moving configurations between environments

Remote Profiles and Perimeter Servers are exportable resources, just like Business Processes, Maps, and Trading Partner Data — Deployment > Resource Manager > Import/Export. This is how a Remote Profile built and tested in a training/dev environment moves cleanly into test, staging, or production without being rebuilt by hand. Both environments should be on the same product version to limit import/export issues. A bundled .jar export can carry Trading Envelopes, XSLT Style Sheets, Maps, PGP Profiles, Perimeter Servers, and more in a single package.

Adapters and considerations

- SFTP Client Adapter — the primary adapter for Put/Get/mput/mget operations against a partner endpoint using a named Remote Profile; the adapter used in this lesson's exercises.
- Command Line 2 Adapter — a scripted alternative for moving files, already seen in the Vendor Managed Inventory capstone (copytest.sh / copytest.bat). Useful when file-movement logic is easier to express as a script than as adapter configuration.
- Perimeter Server — in real enterprise deployments, discuss this as a real-world consideration for any partner sitting outside your firewall or DMZ, even if your lab doesn't have one configured.

Security best practices

- Disable password authentication once key-based authentication is verified working.
- Rotate SSH keys on a defined schedule, and immediately on any suspected compromise.
- Restrict file permissions on private keys at the operating-system level — never share a private key file.
- Monitor and alert on repeated failed authentication attempts (security.log / common3splogger.log).
- Prefer a Perimeter Server over opening inbound firewall ports directly to the application server.
- Re-verify a partner's Known Host Key out of band whenever it changes — never blindly re-trust a changed key.
- Export and version-control Remote Profile / Perimeter Server resources as part of your change-management process.

Source: Advanced Business Process Modeling (6F100G) — Managing Resources / Import-Export resource types (including Perimeter Servers and PGP Profiles).

Quiz

1. What is the key difference between a Remote Profile and hard-coding connection details directly into a business process?
2. Name the four SSH keys used in Sterling B2B Integrator and state, for each, whether it identifies your organization or your trading partner.
3. Why must a client verify a server's host key before trusting a connection?
4. What problem does a Perimeter Server solve, and where is it physically deployed?
5. List three fields you must configure on an SSH Remote Profile, beyond the host and port.
6. Which SFTP Client service would you use to remove a file from the remote server after a successful GET, and why is that a good practice?
7. A partner reports "host key not accepted" errors starting today. What are two possible explanations, and which log would you check first?
8. Name two ways a Remote Profile configured in a training environment could be moved into a production environment without being rebuilt by hand.

Lesson Review

In this lesson, you learned why SSH/SFTP is the standard for secure trading-partner file transfer and where it sits among the other protocols Sterling B2B Integrator supports; how the SSH handshake and the underlying symmetric/asymmetric cryptography work; the four SSH keys Sterling B2B Integrator uses and where each is configured; the role of a Perimeter Server in a DMZ deployment; how to configure a Remote Profile with both password and key-based authentication using the real product field set; how to build and run a business process that moves a file through that profile using the SFTP Client services; and how to diagnose a failed secure connection using the correct log files. The next session moves from connectivity into operational readiness: Scheduling and the Health Check Utility.

Appendix: Source References

Uploaded course guides used

- IBM Sterling B2B Integrator Administration and Tuning V5.2.6.1 (6F89G) — Perimeter Servers, Architecture and Components, Troubleshoot System Logs, FTP Server Adapter exercises (4.2.1 / 4.2.2).
- IBM Sterling B2B Integrator Fundamentals (6F870GuideV2) — Deployment menu descriptions including SSH Host Identity Key.
- Managing Sterling File Gateway V2.6.6.1 (6F86G) — supported partner communication protocols, SFTP/SSH profile prerequisite for consumer partners.
- Advanced Business Process Modeling (6F100G) — FTP/SFTP Client Services, Managing Resources / Import-Export resource types.

Public-domain / IBM documentation used

- IBM Support — "The difference between the 4 SFTP (SSH) keys in IBM Sterling B2B Integrator."
- IBM Sterling B2B Integrator product documentation — SSH Remote Profile configuration field reference (Trading Partners > SSH > Remote Profiles) and Known Host Key check-in methods.
- IBM community reference material — SSH/SFTP key-based authentication setup procedure in Sterling B2B Integrator.
- General SSH protocol references — the SSH handshake sequence and symmetric-vs-asymmetric cryptography are established, publicly documented security-industry concepts and are not specific to any single source.

Note on the pilot version

An earlier draft of this lesson was built primarily from general Sterling B2B Integrator domain knowledge because no dedicated "SSH & Remote Profiles" lesson exists in any of the 20 IBM course guides supplied for this course. This revision replaces that general-knowledge draft with the sourced content above wherever a direct citation was found; the SSH handshake and cryptography primer remain general-knowledge sections by nature, and are labeled as such in the Instructor Note at the start of this guide.